

Q1. RIP — Chennai HQ Campus

a) Technical Summary & Trigger: NOC changes CHG-0221 and CHG-0224 introduced a temporary VLAN gateway chain for Delivery department onboarding, placing the new subnet 16 router hops away from the core switch. The route was silently omitted, leaving client services offline until tickets were opened.

b) Hop-Count Execution Rules: RIP v2 uses hop count as its sole metric with a maximum limit of 15. The 16th hop boundary evaluated the metric as 16 (Infinity/Unreachable). Following distance-vector rules, RIP v2 silently dropped the subnet from routing update advertisements and omitted it from the active forwarding table.

c) Fault Analysis & Monitoring Gap: RIP v2 produces no error logs or alerts when routes exceed 15 hops; it suppresses advertisements by design. NMS tools only tracked physical interface states (UP/DOWN), creating a gap where healthy physical links masked a silent control-plane routing failure.

d) Solution & Historical Justification: Flatten the campus network using VLSM directly at the core/distribution layer or migrate campus routing to single-area OSPF (Area 0). This is highly feasible because Chennai HQ hosts the Area 0 core router. RIP v2 was originally a reasonable choice due to the campus's small, stable topology (Finance/HR) before the onboarding expansion.

Q2. IGRP — Bengaluru–Pune–Hyderabad Regional Link

a) Case Summary: An ISP fault degraded the direct 2 Mbps Bengaluru–Pune link, causing latency spikes (600ms+) for client applications. Although a healthier 8 Mbps alternate path via Pune–Hyderabad–Bengaluru existed, traffic remained trapped on the degraded direct link.

b) Composite Metric Formula Calculation: IGRP uses $\text{Metric} = \text{Bandwidth_Scaled} + \text{Delay_Scaled}$ under default settings. The degraded 2 Mbps direct link's high latency surged its metric value, while the 8 Mbps alternate path produced a significantly lower numerical metric, mathematically defining the alternate path as preferred.

c) Reconvergence Lag Analysis: This was a convergence-speed issue, not a metric calculation failure. IGRP calculated updated metrics correctly, but its legacy periodic update timers (90s updates, holddown timers) delayed propagation across the network, keeping traffic on the degraded link.

d) Solution & Protocol Justification: Decommission IGRP and integrate regional links into multi-area OSPF (Areas 1, 2, and 3). IGRP was originally chosen to overcome RIP's 15-hop limit via composite metrics, but failed under real link degradation due to slow timer-based convergence.

Q3. OSPF — Core Backbone Migration

a) Cause & Routing Black Hole: Change CHG-0245 entered an incorrect Area ID on the Hyderabad router during OSPF migration. While Layer 1/2 physical links stayed UP, control-plane neighbor adjacency failed, causing the router to silently drop incoming packets.

c) Operational Gap Analysis: Control-plane misconfigurations leave physical links UP, bypassing standard SNMP interface polling. Organizational gaps included lack of configuration peer review,

absence of mandatory CLI verification checks (`show ip ospf neighbor`), and concurrent migration/onboarding changes without a freeze window.

d) Solution & Multi-Area OSPF Benefits: Correct Area ID configuration and enforce two-person peer authorization with automated post-implementation CLI scripts. Multi-area OSPF bounds LSAs to regional areas, provides sub-second Dijkstra SPF convergence, and eliminates distance-vector loop vulnerabilities.

Q4. BGP — ISP Peering and Failover

a) Failure Event & Intervention: On March 18 at 14:50 IST, primary ISP-A suffered a fiber cut. Automatic failover to secondary ISP-B failed, stalling outbound traffic until senior architects executed manual BGP overrides at 15:40 IST to force route selection toward ISP-B.

b) BGP Path Attribute Analysis: Outbound preference was dictated by `LOCAL\_PREF` (ISP-A set to 200 vs ISP-B set to 100). Lacking dynamic SLA tracking or interface teardown triggers, BGP kept preferring the dead ISP-A path despite upstream link failure.

c) Risk & Business Consequence Chain: BGP relies on explicit administrative policies across Autonomous Systems. Leaving failover policies untested since October 2023 caused an extended 9-hour 13-minute outage, directly breaching contractual 99.9% uptime SLAs for 6 enterprise clients.

d) Solution & Architectural Justification: Bind BGP `LOCAL\_PREF` assignments to automated ICMP IP SLA tracking probes and establish quarterly failover test windows. BGP is essential at the WAN edge because IGP cannot evaluate external Autonomous System path attributes or manage multi-homed ISP policies.